Speech Emotion Recognition for the Jordanian Arabic Dialect

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Abstract

Speech Emotion Recognition (SER) has been receiving increasing attention in recent years due to its applications in human-computer interaction (HCI) and affective computing. Nevertheless, there is limited research on speech emotion recognition in Arabic dialects. Additionally, there are very few studies that focus on the Jordanian dialect. In this paper, a speech emotion recognition system is proposed based on a recently recorded dataset comprising 1,217 speech samples in the Jordanian Arabic dialect. Four emotions: happy, sad, angry, and neutral are considered in this study. Additionally, the dataset includes 12 different speakers. The dataset was recorded in a controlled environment as well as through social media platforms to obtain voice samples from different individuals. In this study, several methods are considered, including conventional machine learning classifiers, SVM, KNN, and MLP, after performing feature engineering in the form of Mel Frequency Cepstral Coefficients (MFCC). Additionally, a deep learning-based transfer learning approach utilizing the Facebook Wav2Vec 2.0 model is employed. Additionally, a hybrid feature fusion approach based on MFCC and Wav2Vec 2.0 is considered. Experimental results show that although the SVM classifier provided an accuracy of 86.41%, the Wav2Vec 2.0 model alone achieved a comparable result of 87.50%. However, the Combined Model (MFCC + Wav2Vec 2.0) outperformed individual models and achieved a maximum accuracy of 90.22%. These outcomes highlight the benefits of feature fusion and transfer learning for effective SER systems in under-resourced Arabic dialects.

CCS Concepts

• **Computing methodologies** → **Speech recognition**; *Supervised learning by classification.*

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Keywords

Speech Emotion Recognition, Jordanian Arabic Dialect, Arabic speech, SVM, Transfer Learning, Wav2Vec 2.0, MFCC, KNN, MLP, Hybrid Model

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1 Introduction

Speech is a natural form of communication that is both specific and expressive and can be described as "the use of oral media in passing information from a speaker to an audience, whether formally or informally." This information can range from the speaker's ideas, knowledge, and feelings, which can help in the expression of thoughts and feelings, building relationships, and sharing emotions. Though human beings can easily comprehend emotions, there is still a challenge in getting machines to perceive and interpret human emotions.

There has been ongoing interest in emotion recognition by machines due to advancements in Artificial Intelligence (AI) and Machine Learning (ML), particularly Deep Learning (DL). In recent years, the computational theory of speech emotion recognition has been studied extensively [7]. It refers to the process through which machines can identify and classify emotions such as happiness, sadness, anger, or neutral speech signals. In addition, it analyzes linguistic features such as words used, tone, speed, and intensity.

Fig. 1 shows the major applications areas of SER, including Human-Computer interaction, healthcare and mental health, customer services, education and e-learning, and business and market research. These applications enhance interactions, decision-making and effectiveness in various fields.

Regarding the languages, most of the datasets for speech emotion are implemented in German languages [5, 17], English, and Spanish. There are also some multi-language datasets such as Ryerson Multimedia Lab (RML), which comprises languages such as

Figure 1: Major applications areas of SER.

English, Italian, Mandarin, and Urdu languages [4]. A few studies have also been conducted on Indian languages [2]. However, though there are a few studies, they are still not adequate, despite the fact that it is one of the most widely spoken languages and is the official language of many nations [13]. This limitation has also been highlighted in recent systematic reviews of speech emotion recognition research [8].

The formal language used in Arabic is called Modern Standard Arabic, but it can vary depending on geographic distribution. There are many datasets developed for Tunisia [9] and Saudi Arabia [1]. In this paper, we focus on creating a dataset of Jordanian dialect that is easily understandable, especially for the Levant region, and addresses a research gap.

The datasets include four commonly evaluated emotions: anger, happiness, sadness, and neutral. To extract features, MFCCs were used to obtain spectral features that are useful for the models. The most widely used machine learning classification methods were then applied to SER, such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Multi-Layer Perceptron (MLP). Wav2Vec2 was also used to extract deep features from raw audio. In addition, MFCCs and Wav2Vec2 features were combined to improve emotion classification, which has been shown to achieve good performance.

The outline of this paper is as follows: In Section 2, the related work in this area of study is discussed. In Section 3, the methodology of this study is explained. In Section 4, the experiments conducted in this study and the models used in this study are discussed. In Section 5, the results of this study are presented and analyzed. In Section 6, the results obtained in this study and the errors associated with this study are discussed. Finally, in Section 7, the conclusion of this study and the directions for future study in this area of study are discussed.

2 Related Work

Previous studies in Speech Emotion Recognition (SER) mainly focused on using handcrafted acoustic features like Mel-frequency cepstral coefficients (MFCC) and using traditional classifiers for the same [6]. For instance, Milton et al. [17] introduced a hierarchical Support Vector Machine (SVM) classifier on the Berlin Emotional Database with a low accuracy level of 68%. However, SVMs have indicated their potential in controlled environments. Chavhan et al. [5] worked with the same Berlin database but obtained remarkable accuracy of 96.25% using a polynomial kernel with Mel Energy Spectrum Dynamic Coefficients (MEDC). Even though the results are impressive, it has indicated the major drawback of the traditional model in handling clean and speakable data.

This was further investigated in a study carried out by Jain et al. [11]. The study found a wide gap in performance between SVM's 90.08% accuracy for acted speech (LDC dataset) and a drastic fall to 65.97% for elicited spontaneous speech (UGA dataset). The fall in performance highlights the limitation of hand-engineered features in effectively generalizing to real-world communication scenarios and hence the need for a paradigm shift to DL-based features.

In addition to SVMs, other traditional machine learning algorithms have played a crucial role as a necessary benchmarking tool in SER studies. Meftah et al. [15], for example, focused on the application of the K-Nearest Neighbors (KNN) classifier on the KSUEmotions corpus. On the other hand, in relation to neural network baselines, Aljuhani et al. [3], Ismaiel et al. [10], and Joy et al. [12] applied Multi Layer Perceptrons (MLP) as a basic neural network strategy for the classification.

The advent of Deep Learning techniques like CNNs and RNNs signified the first instance of automatic representation learning. To enhance these features extracted using conventional methods, Aouani and Ben Ayed proposed the Stacked Auto-Encoder (AE) to reduce the dimensionality of acoustic features like MFCCs and HNR in [4]. The hybrid model using an SVM classifier attained an accuracy of 74.07% using the RML dataset, thus indicating that feature selection using neural networks produces better feature representation than conventional features.

However, to address the challenges of modeling long range dependencies, the attention shifted to the Transformers. Inspired by this, Akinpelu et al. [1] successfully applied Vision Transformers to spectrograms, resulting in state-of-the-art accuracy. To complement this, Pepino et al. [18] utilized Wav2Vec 2.0, proving that aggregating representations from multiple layers is more effective in capturing paralinguistic information compared to the standard ASR fine-tuning.

Though there has been a lot of advancement in the field of SER globally, research in the context of the Arabic language is restricted because there are no publicly available datasets. Aljuhani et al. [3] have created a semi-natural dataset for the Saudi dialect from YouTube videos. Using the extracted spectral features MFCCs and Mel-spectrograms, they have tested several classifiers. SVM performed best in terms of accuracy at 77.14% in comparison to KNN and MLP classifiers. Ismaiel et al. [10] have utilized the Arabic Natural Audio Dataset (ANAD) to compare the performance of conventional classifiers with deep learning classifiers. The Deep Neural Network (SER-DNN) classifier has been proposed in this research. This classifier has been found to have a higher accuracy of 97.40% on three classes in comparison to ensemble classifiers.

Moreover, working with multi-dialect data also poses additional challenges. Meftah et al. [15, 16] worked on the KSUEmotions dataset, which includes recordings from Saudi, Syrian, and Yemeni speakers. Using the Residual Neural Network (ResNet), the researchers obtained an accuracy of 85.53% on five emotion classes. What is interesting to notice is the fact that the model failed to

recognize positive emotions such as happiness with the same precision as negative emotions such as anger, thereby emphasizing the necessity to use robust representations such as Wav2Vec 2.0.

Ibn Nasr et al. [9] used classical machine learning methods on spontaneous spoken Tunisian dialect collected from YouTube. The maximum accuracy obtained was 58.75%, which indicates that classical machine learning methods are not efficient for spontaneous spoken dialects. In addition, unlike previous works on Arabic SER, this study focuses on feature-level fusion of handcrafted acoustic features and self-supervised speech representations on the Jordanian dialect.

3 Methodology

3.1 Dataset

3.1.1 Data Collection. We also created our own Jordanian Arabic Speech Emotion Recognition dataset from scratch using standard procedures in speech emotion recognition methodology [14]. The procedures involved the following steps: First, we created 300 sentences for each of the four emotions: happy, sad, angry, and neutral. The sentences were created in such a way that they naturally carry each emotion. We then recruited 12 native Jordanian Arabic speakers, 10 males, and 2 females, to record the dataset. Each person was given the sentences to speak while carrying out the emotion naturally. The recording was done using smartphone microphones in normal environments without any special recording studio equipment.

After recording, we performed manual verification by listening to all audio samples to ensure each recording correctly represented its intended emotion. Samples that did not clearly express the target emotion were removed or re-recorded. Finally, the dataset was shuffled and split into training (70%), validation (15%), and testing (15%) sets.

3.1.2 Dataset Description. The final dataset contains 1,217 audio samples across four emotion classes. Each audio recording is 2–5 seconds in length. Table 1 shows the distribution of samples across emotions and data splits.

Table 1: Dataset Distribution

Emotion	Train	Val	Test	Total
Happy	226	49	49	324
Angry	212	46	46	304
Neutral	209	45	45	299
Sad	202	44	44	290
Total	849	184	184	1,217

3.2 Preprocessing

Firstly, the audio files are converted to a single channel audio file. This is done to reduce the file size. Also, it is required by the Wav2Vec2 model. Then, the audio files are resampled at a frequency of 16,000 Hz. This is required by the Wav2Vec2 model. The duration of the audio file (2–5 seconds) is kept the same. Finally, the volume is normalized to the range $[-1, 1]$.

3.3 Feature Extraction

3.3.1 MFCC Features. Mel Frequency Cepstral Coefficients (MFCCs) are extracted as conventional acoustic features since they have been widely applied in speech emotion recognition systems due to their successful usage in modeling human auditory perception [19], utilizing the Librosa library. For each audio recording, 40 MFCCs are calculated. Then, statistical values such as mean, standard deviation, minimum, and maximum are calculated for each MFCC. In addition, delta MFCCs are extracted as first-order derivatives. Finally, the mean and standard deviation values are calculated. Furthermore, 12 chroma features, spectral centroid, zero-crossing rate, and root mean square (RMS) energy are extracted. In total, 258 features are extracted from each audio recording.

3.3.2 Wav2Vec2 Embeddings. For the features extracted using deep learning, we employed the pre-trained Wav2Vec2-Large-XLSR-53 model from the Hugging Face Transformers library. This model was trained on 56,000 hours of speech samples in 53 languages, including Arabic, making it relevant for our Jordanian dialect dataset. The model was employed as a frozen feature extractor without any fine-tuning. For each audio sample, the hidden states of the final transformer layer were extracted, and mean pooling was performed along the time axis to extract a fixed-size vector. This produced a 1,024-dimensional embedding vector for each audio sample.

Table 2: Feature Extraction Summary

Feature Type	Method	Dimensions
MFCCs	Librosa	258
Wav2Vec2	Pretrained XLSR-53	1,024
Combined	MFCC + Wav2Vec2	1,282

3.4 Models

3.4.1 Wav2Vec2 Classifier. The Wav2Vec2 embeddings (dimension 1,024) were normalized first using StandardScaler. The output of this step was fed to a neural network classifier with three fully connected layers. The structure was as follows: Linear($1024 \rightarrow 512$) followed by BatchNorm, ReLU activation, and Dropout (0.3), then Linear($512 \rightarrow 256$) with the same structure, and finally Linear($256 \rightarrow 4$) for the output layer with four classes for emotion.

3.4.2 SVM. As the baseline model, we employed Support Vector Machine (SVM) with MFCC features (258 dimensions). The features were normalized by StandardScaler prior to training. The kernel function used is RBF (Radial Basis Function), which is the most popular kernel function for SVM since it can deal with non-linearly separable data by introducing decision boundaries with curved shapes. The hyperparameters were fixed at $C = 10$, which regulates the hardness of the decision boundary, and γ = 'scale', which adjusts the weight of each sample automatically. The selection of SVM is

justified by its success in various speech emotion classification problems [20].

3.5 Training

For training the neural network models (Wav2Vec2 classifier and Combined model), we utilized the AdamW optimizer with a learning rate of 0.0001 and weight decay of 0.01. A CosineAnnealingLR scheduler was used to decrease the learning rate during training. The models were trained for 100 epochs with a batch size of 16, using Cross-Entropy Loss as the loss function. To avoid overfitting, early stopping was implemented by saving the model with the highest validation accuracy. The dataset was divided into training (70%), validation (15%), and testing (15%) sets.

4 Experimental Setup

4.1 Dataset and Preprocessing

As per the conventional evaluation protocols, the dataset has been divided into three sets: 70% for training, 15% for validation, and 15% for testing. In addition, since the architecture of the pre-trained model is based on Wav2Vec 2.0, it has been resampled at a frequency of 16 kHz. Also, the audio waveforms have been normalized to the range $[-1, 1]$.

4.2 Baselines and Comparison Models

In order to validate the efficacy of our proposed deep learning technique, we compare it with strong baseline models defined in the SER literature. These include Support Vector Machine (SVM) with an RBF kernel function, where the regularization parameter (C) and kernel coefficient (γ) are optimized on the validation set through grid search; K-Nearest Neighbors (KNN) classifier with different values of neighbors (k) to determine the optimal local structure; and Multi-Layer Perceptron (MLP), a strong baseline for neural networks comprising three hidden layers with 256, 128, and 64 units each, utilizing ReLU as an activation function to bridge shallow and deep learning.

4.3 Implementation and Hyperparameters

All experiments were conducted using the PyTorch framework on a workstation equipped with an NVIDIA GPU. Deep learning models (Wav2Vec 2.0 and the Combined Model) were trained for 100 epochs with a batch size of 16. We minimized the Cross-Entropy Loss using the AdamW optimizer with a learning rate of 0.0001 and a weight decay of 0.01. A Cosine Annealing scheduler was employed to dynamically adjust the learning rate during training. The best model checkpoint was saved based on the highest accuracy achieved on the validation set.

4.4 Evaluation Metrics

The performance of the models was thoroughly assessed using Accuracy as the principal global metric. To gain further insights into the performance of the models on a per-class basis, especially in the context of distinguishing among certain emotions, Precision, Recall, and F1-score were also calculated. Further, Confusion Matrices were drawn to analyze the patterns of misclassification among the four classes of emotions (Happy, Sad, Angry, and Neutral) and to understand the certain limitations in the ability to recognize the nuances of the Jordanian dialect.

5 Results and Analysis

5.1 Overview

This section will introduce the results of the experiments conducted on the proposed models using the Jordanian Arabic dataset. The classification accuracy of the proposed models is calculated using the classification accuracy metric, as shown in Table 3.

5.2 Support Vector Machine (SVM) Results

Support Vector Machine (SVM) was used to evaluate the performance of handcrafted MFCC features. The SVM classifier obtained a training accuracy close to perfection: 99.88%. The result indicates that the RBF kernel function successfully mapped the training samples. However, it was also noted that there was a generalization gap since the validation and test set accuracies were 82.07% and 86.41%, respectively. The difference in training and testing accuracy indicates that there was some level of overfitting in the system. The overfitting could be attributed to the small dataset used for training compared to the dimensionality of the feature space. The test set accuracy of 86.41% indicates that it is possible to solve dialectal SER problems despite the constraints.

5.3 K-Nearest Neighbors (KNN) Results

K-Nearest Neighbors (KNN) model was assessed using MFCC features with $k = 1$. Although the model had a near-perfect training accuracy of 99.88%, its performance on new data was not impressive, with the model attaining a validation accuracy of 75.00% and a test accuracy of 78.80%. This shows a reasonable ability to classify emotions, but the difference in training and test accuracy shows that KNN heavily relies on memorization rather than learning.

5.4 Multi-Layer Perceptron (MLP) Results

The Multi-Layer Perceptron (MLP) model, which was trained on MFCC features to identify non-linear patterns, resulted in a training accuracy of 99.88%. The MLP model showed better generalization capabilities than the KNN model, as it reached a validation accuracy of 78.26% and a test accuracy of 79.89%. Although the MLP model marginally outperformed the instance-based KNN classifier, it failed to beat the SVM baseline accuracy of 86.41%, suggesting that although neural networks are very capable, simple models with small datasets may not be able to generalize as well as kernel-based models such as SVM in high-dimensional spaces.

5.5 Wav2Vec2-Based Model Results

The Wav2Vec2-based model performed well by utilizing pre-trained representations of speech. The model attained a training accuracy of 99.88%, a validation accuracy of 87.50%, and a test accuracy of 87.50%. These values prove the success of self-supervised learning in speech emotion recognition, as the model generalizes better than the traditional MFCC-based methods.

Table 3: Performance comparison of the evaluated models in terms of Training, Validation, and Test accuracy.

Model	Feature Set	Training Acc. (%)	Validation Acc. (%)	Test Acc. (%)
KNN (k=1)	MFCCs	99.88	75.00	78.80
MLP	MFCCs	99.88	78.26	79.89
SVM (Baseline)	MFCCs	99.88	82.07	86.41
Wav2Vec 2.0	Deep Embeddings	99.88	87.50	87.50
Combined Model	Fusion (MFCC + Wav2Vec)	99.88	87.50	90.22

5.6 Combined MFCC + Wav2Vec2 Model Results

The best overall performance was achieved by the combined model that fused MFCC features with Wav2Vec2 embeddings. The model attained a training accuracy of 99.88%, a validation accuracy of 87.50%, and a test accuracy of 90.22%. The above-mentioned results clearly indicate that the feature-level fusion technique is capable of effectively capturing complementary information from both hand-crafted acoustic features and learned speech representations.

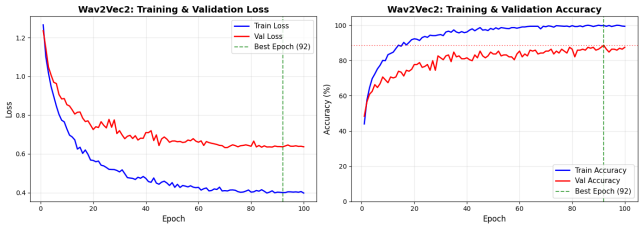

Figure 2: Training and validation accuracy curves of the Wav2Vec2-based transfer learning model.

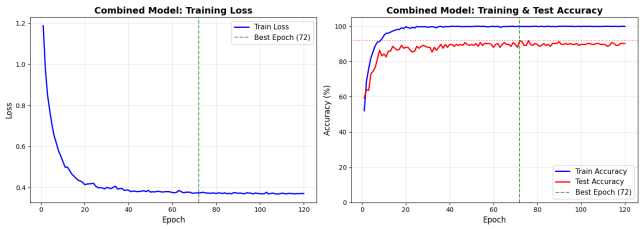

Figure 3: Training and validation accuracy curves of the Combined MFCC + Wav2Vec2 model.

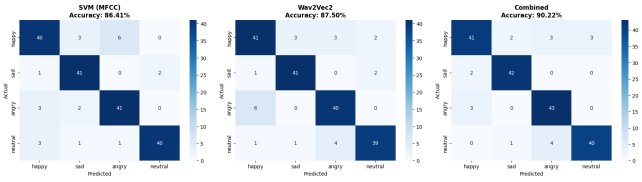

Figure 4: Confusion matrices for the SVM, Wav2Vec2, and combined MFCC–Wav2Vec2 models.

6 Discussion and Error Analysis

6.1 Training Accuracy Observation

All the models in the evaluation had a near perfect training accuracy, with a percentage of 99.88%. This can primarily be attributed to the small size of the dataset and the recording setup, which is favorable for fitting the training data for the high capacity models. There is a consistent performance degradation in the validation and test sets for all the models, suggesting that this is a case of mild overfitting rather than data leakage.

6.2 Confusion Matrix and Error Analysis

In addition to the confusion matrices shown in Fig. 4, the classification reports (detailed in Table 4) provide deeper insight into class-wise performance.

Table 4: Detailed performance comparison (Precision, Recall, and F1-score) for each emotion class across the three models.

Model	Emotion	Precision	Recall	F1-Score
SVM	Happy	0.78	0.82	0.80
3*(Baseline)	Sad	0.91	0.91	0.91
	Angry	0.83	0.87	0.85
	Neutral	0.95	0.87	0.91
4*Wav2Vec 2.0	Happy	0.84	0.84	0.84
	Sad	0.91	0.93	0.92
	Angry	0.85	0.87	0.86
	Neutral	0.91	0.87	0.89
Combined Model	Happy	0.89	0.84	0.86
3*(Proposed)	Sad	0.93	0.95	0.94
	Angry	0.86	0.93	0.90
	Neutral	0.93	0.89	0.91

The precision and recall of the SVM model for the 'Happy' class are relatively lower (F1-score = 0.80), implying that there is a tendency to misclassify the speech as neutral. This is an indication that the MFCC features are not sufficient to distinguish the subtle expression of emotions, especially when happiness is expressed at low intensity.

It is noted that the Wav2Vec 2.0-based model shows better performance in all emotion classes, especially for 'Sad' and 'Angry'

emotions, where it has achieved better recall values, namely 0.93 and 0.87, respectively. This is because the model uses contextual and paralinguistic information, which is not present in spectral features.

Lastly, the Combined MFCC-Wav2Vec 2.0 model shows the most balanced performance in terms of macro-averaged precision, recall, and F1-score, all of which are 0.90. Specifically, it minimizes the confusion between emotions with similar characteristics, like 'Happy' and 'Neutral.' Moreover, it shows the highest recall for 'Sad' (0.95) and 'Angry' (0.93) emotions. This shows that the use of feature fusion indeed increases robustness in the discrimination of emotions for the Jordanian Arabic dialect.

6.3 Limitations

Despite promising results, this study has limitations that should be acknowledged.

- **Dataset Size:** Although our database of 1,217 samples is adequate for baseline testing, it is still limited in comparison to the large datasets used in high-resource languages that could affect the ability of the deep learning model to generalize the unseen speakers perfectly. In an initial attempt to address the limited size of our database and avoid overfitting, we also attempted to apply some conventional audio data augmentation methods like pitch variation and time stretching. However, there was an observable deterioration in the performance of the model. We believe this is because these transformations alter the nuanced paralinguistic features that are characteristic of emotional representation in the Jordanian dialect. We therefore decided to train the model using only the original audio files.
- **Gender Imbalance:** In the current dataset, a gender difference has been observed. There are 10 male speakers in the dataset, while there are only 2 female speakers. This could inherently cause a bias in the model towards male speech characteristics, which could influence the accuracy of the model in recognizing speech. In the coming phases, a strict emphasis will be laid on ensuring a balanced dataset.
- **Speaker-Independent Evaluation Limitation:** The current experimental environment does not require a strictly speaker-independent evaluation protocol, as audio examples of the same speaker can occur in the training, validation, and test datasets. Although this is a common experimental environment for initial SER research, it could potentially result in overly optimistic performance estimates due to speaker-specific information. Future research will incorporate speaker-independent evaluation protocols, such as leave-one-speaker-out cross-validation.

7 Conclusion and Future Work

The paper proposed a speech emotion recognition system for the Jordanian Arabic dialect, which is considered one of the low-resource dialects that have not been adequately addressed in the existing SER literature. The paper also examined the effectiveness of several machine learning and deep learning architectures using MFCCs features, Wav2Vec 2.0 features, and their fusion for the proposed system. The results demonstrated that the proposed system using MFCCs features with an SVM classifier achieved promising results. The proposed system using Wav2Vec 2.0 features achieved improved results due to the ability of this model to generalize better using pretrained speech features. The proposed system using the MFCCs-Wav2Vec 2.0 feature fusion achieved the best results in terms of test accuracy, achieving 90.22%.

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